



DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING

USN No:

FN05650453

II Semester B. Tech. Mid Semester Examination Computer Science and Engineering

Course Title: C Programming for Problem Solving (CPPS)

Course Code: 25EN1114

Duration: 75 Min.

Date: 14-03-2026

Time: 1.00 to 2.15 PM

Max Marks: 40

Instructions to Candidates:

- Answer all the four Questions
- Each question carries 10 marks

		Marks	BT Level	CO's	PO's
Q. I(a)	ii. Which loop is guaranteed to execute at least once? A) for loop B) while loop C) do-while loop D) None iv. Which operator is to find the remainder of division? A) / B) % C) * D) & iii. What is the value of x in this expression $X=10+5*2$? A) 30 B) 20 C) 25 D) 15 v. Which function is used to compare two strings in C? A) cmpstr() B) compare() C) strcmp() D) strcomp() i. Which of the following is a valid identifier in C? A) my_variable1 B) 2ndVariable C) float D) name!	05	2	1, 2	1
Q. I(b)	Predict the output of the following code snippet? <pre>#include<stdio.h> int main() { int i; for (i=1; i<=5; i++) { if(i == 3) { break; } printf("%d ", i); } return 0; }</pre> 265 1, 2, 3 <pre>#include<stdio.h> int main() { int x = 4; switch(x) { case 1: printf("One"); break; case 2: printf("Two"); break; default: printf("Other"); } return 0; }</pre> other	05	3	1	2

Q. 2(a)	Construct a C program to generate the roots of a quadratic equation. The various parameters for the quadratic equation are to be received as inputs from the user.	05	2	1	3
Q. 2(b)	Evaluate the following arithmetic and conditional C expressions, when a=2, b=4, c=3: iii. $Z = c * a \% b + 1$ iv. $X = (a + b) > c \ \&\& \ a < b$	05	3	1	2
Q. 3(a)	Develop a C program for the scenario, where A teacher wants to analyse the performance of students in a recent exam. The scores of 10 students are stored in a 1D array. The teacher wants to the Count of how many students scored above 75. Also Find the average score.	05	4	2	3
Q. 3(b)	Predict the output of the following code snippet? <pre>#include <stdio.h> int main() { int arr[] = {3, 6, 9, 12}; for (int i = 3; i >= 0; i--) printf("%d ", arr[i]); return 0; }</pre> 12 9 6 3 <pre>#include <stdio.h> #include <string.h> int main() { char str[] = "Data"; printf("%d", sizeof(str)); printf("%d", strlen(str)); return 0; }</pre> 5 4	05	3	2	2
Q. 4(a)	Illustrate the following string handling built-in functions and display the output. i. strcpy() ii. strcat() iii. strlen() iv. strcmp() v. strchr()	05	3	2	3
Q. 4(b)	Develop a C program to implement the addition of two matrices.	05	4	2	3

$$40 \times 5 = 200$$

$$40 + 40 + 36 + 35$$

$$160 + 40 = 200$$



SCHOOL OF ENGINEERING

USN: **EN02SC90U53**

II Semester B.Tech Mid Semester Examination

Department of Mathematics

Course Title : Single and Multivariate Calculus

Course Code : 25EN1201

Duration : 75 Mins.

Date : 13/03/2026

Time : 9.00 AM - 10.15 AM

Max. Marks : 40

Instructions to Candidates: 1. Answer all the questions. 2. Each full question carries 10 marks.

Q.No.	Question Description	Marks	BT Level	COs
1. a)	Using chain rule, express $\frac{dw}{dt}$ in terms of t , where $w(x, y, z) = x^2 + xy + z^2$, $x = 2t$, $y = t^2$, $z = \sin t$.	4	L2	CO1
1. b)	Find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 - 4x - 2y + 1$, on the triangular region in the first quadrant bounded by $x = 0$, $y = 0$, $y = 4 - x$.	6	L3	CO1
2. a)	Test the continuity of the function at the origin $f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$	4	L2	CO1
2. b)	Applying Lagrange's multiplier method, find the extreme values of the function $f(x, y) = 4xy$ subject to the constraint $\frac{x^2}{8} + \frac{y^2}{16} = 1$.	6	L3	CO1
3. a)	Evaluate the double integral $\iint_R (3x^2 + 2y) dA$, where R is bounded by $0 \leq x \leq 2$, $0 \leq y \leq 2x$.	5	L4	CO2
3. b)	Using polar coordinates, evaluate the integral $\iint_R (x^2 + y^2) dA$, where R is the semicircular region bounded by the x -axis and the curve $y = \sqrt{1 - x^2}$.	5	L4	CO2
4. a)	Solve $\int_0^1 \int_0^{1-x} \int_0^{x-y} dz dy dx$.	5	L3	CO2
4. b)	Evaluate the triple integral in the cylindrical coordinates $\int_0^3 \int_0^{2\pi} \int_0^{\cos\theta} r dr d\theta dz.$	5	L4	CO2